

Gabriel Thien

Graduate Engineer (Embedded Software / Mechatronics) @ Cochlear Ltd.

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[LinkedIn](#)

[GitHub Repositories](#)

SUMMARY

Hungry & curious problem-solver with a enthusiastic attitude towards making things. Loves streamlining processes and making things easier, more advanced and flexible. Engineer with experience in embedded software architecture, microcontrollers RTOS, automation & mechatronic systems, IoT and product development. Team-player who loves tinkering in the workshop, owning and developing end to end solutions from low-level electronics to firmware and electromechanical systems. Open & looking to relocate to Melbourne, Victoria.

EDUCATION

UNSW Sydney – (2021 ~ 2025)

Degree: B. Engineering (Mechatronics)/B. Science (Computer Science)

EXPERIENCE

Graduate Engineer (Embedded Software / Mechatronics)

Cochlear Ltd – (Feb 2026 – Present)

- Rotating across manufacturing and R&D, building deep and varied industry knowledge
- Designing and developing automation systems, embedded software, electronics, and manufacturing test equipment
- Identifying and solving business pipeline problems through technology exploration and infrastructure improvements

Associate Firmware Engineer

Cochlear Ltd. – (Nov 2024 – Nov 2025)

- Enhanced RTOS firmware development by developing stack analysis tools, allowing for granular memory usage view of programs and tasks
- Reduced complexity of in-house ASIC & firmware test systems with gRPC architecture
- Secured highly competitive graduate engineer position for 2026, rotating through various departments

Associate Mechatronics Engineer

Asiga 3D Printers – (March 2023 – Nov 2024)

- Developed bespoke electro-mechanical systems used in manufacturing and quality control
- Designed jigs & fixtures, electronics and firmware bringup
- Developed embedded electronics and control software for motors and sensors

Machine Technician

Archie Brothers Alexandria – (June 2022 – October 2022)

- Improved arcade machine up time with maintenance, fault diagnosis and repairs.

PROJECTS

Logistic & Resource Scheduling System – Factorio (2026)

- Designing a many-to-many, resource, requester, train dynamic optimal scheduling system
- Simulation, nodes and network written in C to Factorio's game mechanics constraints

Common Desktop Controller App for MCUs

- Programming a desktop app that views and controls peripherals attached to MCU
- Displays stream data from MCU, loads new apps and writes memory blocks

QUALITIES

Action-oriented problem solver

Motivated, self-managing workloads and ambitious

Strong collaboration, teamwork, leadership skills

Capable of working independently

Strong communication, presentation, networking skills

Outgoing, friendly, approachable personality

TECHNICAL SKILLS

Embedded systems, FreeRTOS, MCU firmware development in C/C++, Atmega, Espressif, STM32 frameworks

PCB design, electronics circuitry design, TH & SMD soldering, testing & debugging

Design of test systems, fixtures, electronics and processes

3D printing, laser cutting, general workshop skills

C, C# dotnet, C++, Python, Java, SQL, Qt & .NET frameworks, Agile Development, Git

Linux & PLC programming

PC building & installation, desktop support

INTERESTS

Bouldering, hiking, baking

Factorio, mahjong